Лекция 5

17 февраля 2026 г. 9:00

Утв. $f: U_\delta(x_0,y_0) \to \mathbb{R}$, $U_\delta(x_0,y_0) \subset \mathbb{R}^2$

$\exists \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ в $U_\delta(x_0,y_0)$

в (x_0,y_0) $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ непр. в (x_0,y_0)

$\Downarrow$

f диф. в т. (x_0,y_0)

$$f = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Док-во: $f(x,y) - f(x_0,y_0) = f(x,y) - f(x_0,y) + f(x_0,y) - f(x_0,y_0)$

Зафикс. y. $\varphi(x) = f(x,y)$

$\exists \xi: \varphi(x) - \varphi(x_0) = \varphi'(\xi)(x-x_0)$

$f(x,y) - f(x_0,y) = \frac{\partial f}{\partial x}(\xi,y)(x-x_0)$

$f(x,y) - f(x_0,y) = \frac{\partial f}{\partial x}(x_0+\theta,y)(x-x_0)$

$$\frac{\partial f}{\partial x}(x_0+\theta,y) - \frac{\partial f}{\partial x}(x_0,y_0) \xrightarrow[(\theta,y)\to(0,y_0)]{} 0$$

$\left(\frac{\partial f}{\partial x}\text{ непр. в т. }(x_0,y_0)\right)$

$$\left|\frac{\left(\frac{\partial f}{\partial x}(x_0+\theta,y) - \frac{\partial f}{\partial x}(x_0,y_0)\right)(x-x_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}\right| \leq$$

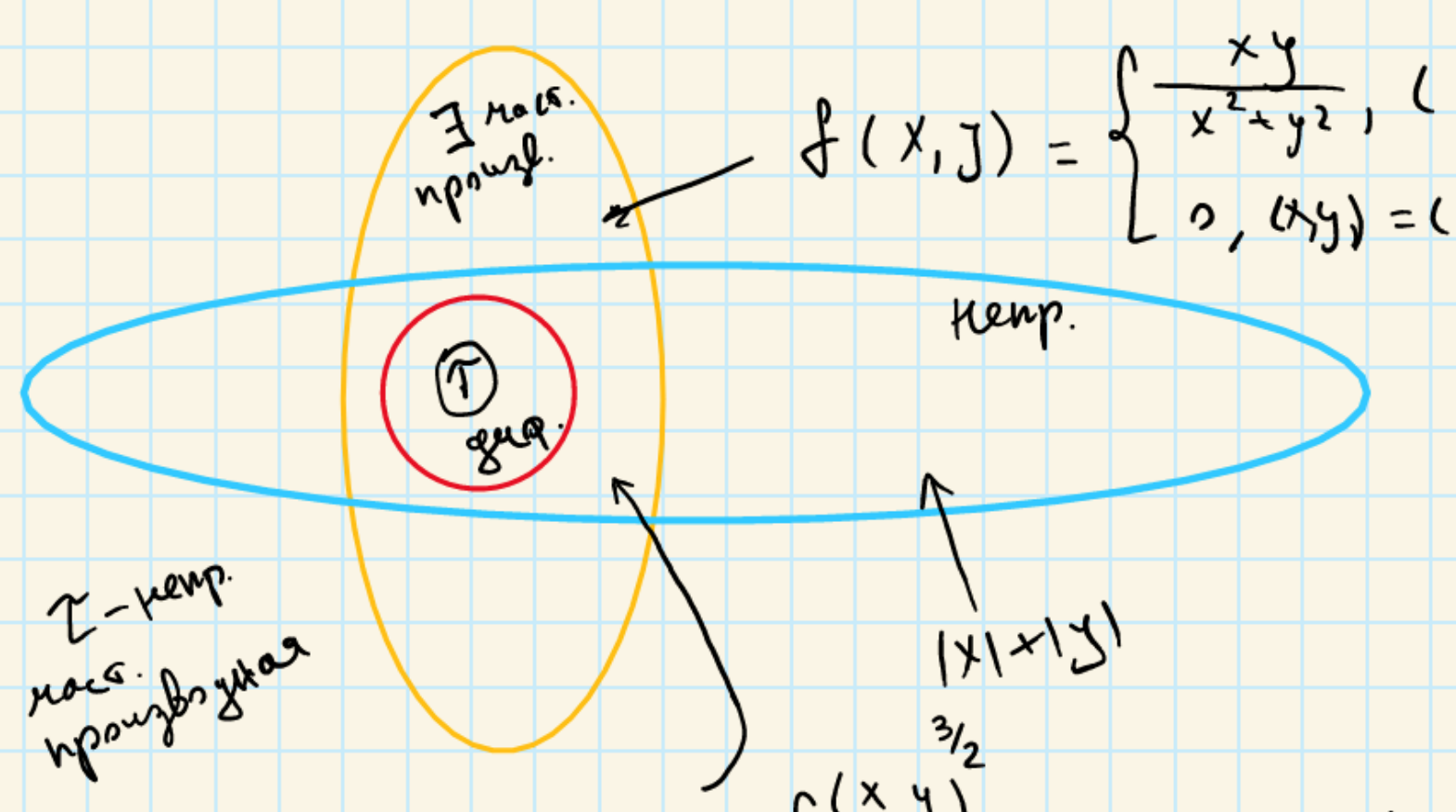
$$\leq \left|\frac{\partial f}{\partial x}(x_0+\theta,y) - \frac{\partial f}{\partial x}(x_0,y_0)\right| \xrightarrow[(\theta,y)\to(0,y_0)]{} 0$$

$\left(\frac{\partial f}{\partial x}(x_0+\theta,y) - \frac{\partial f}{\partial x}(x_0,y_0)\right)(x-x_0) = o(\rho)$. где $\rho = \sqrt{(x-x_0)^2+(y-y_0)^2}$

$$f(x,y) - f(x_0,y) = \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + \underbrace{\left(\frac{\partial f}{\partial x}(x_0+\theta,y) - \frac{\partial f}{\partial x}(x_0,y_0)\right)(x-x_0)}_{o(\rho)\text{ при }\rho\to 0}$$

$f(x,y) - f(x_0,y) = \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + o(\rho)$, $\rho \to 0$

$f(x,y) - f(x_0,y_0) = \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial f}{\partial x}(x_0,y_0)(y-y_0) + o(\rho)$, $\rho \to 0$



$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$f(x,y) = \begin{cases} \frac{(xy)^{3/2}}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

$$\left|\frac{|x|^{3/2}\cdot|y|^{3/2}}{x^2+y^2}\right| = \frac{\rho^3 |\cos\varphi|^{3/2}\cdot|\sin\varphi|^{3/2}}{\rho^2} \leq \rho \xrightarrow[\rho\to+0]{} 0$$

$f(x,0) = 0 \Rightarrow \frac{\partial f}{\partial x}(0,0) = 0$

$f(0,y) = 0 \Rightarrow \frac{\partial f}{\partial y}(0,0) = 0$

$$\frac{f(x,y) - f(x_0,y_0) - \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) - \frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$$

$$\frac{f(x,y)}{\sqrt{x^2+y^2}} = |\cos\varphi|^{\frac{3}{2}}|\sin\varphi|^{\frac{3}{2}}$$

$f(x,y) = (xy)^{\frac{2}{3}}$

$f(x,0) = 0 = f(0,y)$

$\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0$

$$\left|\frac{f(x,y)}{(x^2+y^2)^{1/2}}\right| = \frac{\rho^{4/3}(\cos\varphi\sin\varphi)^{2/3}}{\rho} \leq$$

$$\rho^{\frac{1}{3}} \xrightarrow[\rho\to+0]{} 0$$

$\frac{\partial f}{\partial x} = \frac{y^{\frac{2}{3}}}{x^{\frac{1}{3}}}\cdot\frac{2}{3}$ $(x \neq 0)$

Опр. $f: U_\delta(\bar{x}_0) \to \mathbb{R}^m$, $U_\delta(\bar{x}_0) \subset \mathbb{R}^n$

f диф. т. $\bar{x}_0$, если $\exists A_{m\times n}$

$\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0) = A(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)$ при $\bar{x}\to\bar{x}_0$

$$\bar{o}(|\bar{x}-\bar{x}_0|) = \begin{pmatrix} o(|\bar{x}-\bar{x}_0|) \\ \vdots \\ o(|\bar{x}-\bar{x}_0|) \end{pmatrix} \begin{matrix} -1 \\ \\ -m \end{matrix}$$

Опр. $\bar{f}: U_\delta(\bar{x}_0) \to \mathbb{R}^m$, $U_\delta(\bar{x}_0) \subset \mathbb{R}^n$

$$D_f(\bar{x}_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x^1}(\bar{x}_0) & \cdots & \frac{\partial f_1}{\partial x^n}(\bar{x}_0) \\ \frac{\partial f_2}{\partial x^1}(\bar{x}_0) & \cdots & \vdots \\ \frac{\partial f_m}{\partial x^1}(\bar{x}_0) & \cdots & \frac{\partial f_m}{\partial x^n}(\bar{x}_0) \end{pmatrix}$$

матрица Якоби

Утв. $\bar{f}: U_\delta(\bar{x}_0) \to \mathbb{R}^n$, $U_\delta(\bar{x}_0) \subset \mathbb{R}$

$\bar{f}$ диф. в $\bar{x}_0 \Leftrightarrow \exists D_f(\bar{x}_0)$ и

$\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0) = D_f(\bar{x}_0)(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)$ при $\bar{x}\to\bar{x}_0$

Док-во: $\bar{f}$ диф. в $\bar{x}_0 \Leftrightarrow f_k$ диф. в x_0 $\forall k \in \{1,\dots,m\}$ при $\bar{x}\to x_0$

при этом $\text{grad} f_k = \left(\frac{\partial f_1}{\partial y_1}, \dots \frac{\partial f_n}{\partial x_n}\right)$

$f_k(\bar{x}) - f_k(\bar{x}_0) = \frac{\partial f_k}{\partial x^1}(x^1-x_0^1) + \dots + \frac{\partial f_k}{\partial x^n}(x^n-x_0^n) + o(|\bar{x}-\bar{x}_0|)$

$\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0) = D_f(\bar{x}_0)(\bar{x}-\bar{x}_0) + o(|\bar{x}-\bar{x}_0|)$, $\bar{x}\to\bar{x}_0$

Опр. $\bar{f}: U_\delta(\bar{x}_0) \to \mathbb{R}^m$, $U_\delta(\bar{x}_0) \subset \mathbb{R}^n$

$$d\bar{f} \doteq \begin{pmatrix} df_1 \\ \vdots \\ df_m \end{pmatrix} \qquad d\bar{f} = \begin{pmatrix} (\text{grad} f_1, \bar{x}-\bar{x}_0) \\ \vdots \\ (\text{grad} f_m, \bar{x}-\bar{x}_0) \end{pmatrix}$$

Следствие $\bar{f}: U_\delta(\bar{x}_0) \to \mathbb{R}^m$, $U_\delta(\bar{x}_0) \subset \mathbb{R}^n$

$\bar{f}$ диф. в т. $\bar{x}_0$

то $d\bar{f}(\bar{x}_0) = D_f(\bar{x}_0)(\bar{x}-\bar{x}_0)$

Утв. $f: U_\delta(\bar{x}_0) \to Y$, $U_\delta(\bar{x}_0) \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$

$\bar{y}_0 = \bar{f}(\bar{x}_0) \in \text{int}\, Y$

$\bar{g}: Y \to \mathbb{R}^p$

$\bar{f}$ диф. в т. $\bar{x}_0$

$\bar{g}$ диф. в т. $\bar{y}_0$

$\Rightarrow \bar{h}(\bar{x}) = \bar{g}(\bar{f}(\bar{x}))$ определена в $U(\bar{x}_0)$ и дифференцируема в т. $\bar{x}_0$

$D_h(\bar{x}_0) = D_g(\bar{y}_0)\cdot D_f(\bar{x}_0)$

Док-во: $\bar{y}_0 \in \text{int}\, Y \Rightarrow \exists \varepsilon > 0: U_\varepsilon(\bar{y}_0) \subset Y$

$\bar{f}$ диф. в т. $\bar{x}_0 \Rightarrow \bar{f}$ непр. в т. $\bar{x}_0$

$\exists \delta_0: \forall \bar{x} \in U_{\delta_0}(\bar{x}_0) \hookrightarrow \bar{f}(\bar{x}) \in U_\varepsilon(\bar{y}_0) \subset Y$

$\bar{h}$ определена в $U_{\delta_0}(\bar{x}_0)$

$\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0) = D_f(\bar{x}_0)(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)$

$\bar{g}(\bar{y}) - \bar{g}(\bar{y}_0) = D_g(\bar{y}_0)(\bar{y}-\bar{y}_0) + \bar{o}(|\bar{y}-\bar{y}_0|)$ $\bar{y}\to\bar{y}_0$

$\bar{g}(\bar{f}(\bar{x})) - \bar{g}(\bar{f}(\bar{x}_0)) = D_g(\bar{y}_0)\left(D_f(\bar{x})(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)\right) + \bar{o}(|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|)$

$\bar{h}(\bar{x}) - \bar{h}(\bar{x}_0) = D_g(\bar{y}_0)\cdot D_f(\bar{x}_0)(\bar{x}-\bar{x}_0) + D_g(\bar{y}_0)\cdot\bar{o}(|\bar{x}-\bar{x}_0|) + \bar{o}(|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|)$

$D_g(\bar{y}_0)\cdot\bar{o}(|\bar{x}-\bar{x}_0|) = \bar{o}(|\bar{x}-\bar{x}_0|) \in \mathbb{R}^p$

$$\begin{pmatrix} o(|\bar{x}-\bar{x}_0|) \\ \vdots \\ \end{pmatrix} p\text{ строк}$$

$\bar{o}(|\bar{f}(\bar{x}) - f(\bar{x}_0)|)$

k-я компонента $o(|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|) = o(|\bar{x}-\bar{x}_0|)$

Рассмотрим $\varphi(x) = o(|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|)$

$$\frac{\varphi(x)}{|x-x_0|} = \underbrace{\frac{\varphi(\bar{x})}{|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|}}_{\text{беск. мало}} \cdot \overbrace{\frac{|\bar{f}(\bar{x}) - f(\bar{x}_0)|}{|\bar{x}-\bar{x}_0|}}^{\text{огр.}}$$

$$\frac{|\bar{f}(\bar{x}) - \bar{f}(\bar{x}_0)|}{|\bar{x}-\bar{x}_0|} = \frac{|D_f(x_0)(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)|}{|\bar{x}-\bar{x}_0|} \leq$$

$$\leq \frac{|D_f(x_0)(\bar{x}-\bar{x}_0)|}{|\bar{x}-\bar{x}_0|} + \frac{|\bar{o}(|\bar{x}-\bar{x}_0|)|}{|\bar{x}-\bar{x}_0|}$$

$$\left|\sum_{k=1}^m (\text{grad} f_k, \bar{x}-\bar{x}_0)\right| \leq \sum_{k=1}^m |(\text{grad} f_k(\bar{x}_0), \bar{x}-\bar{x}_0)| \leq$$

$$\leq \left(\sum_{k=1}^m |\text{grad} f_k(\bar{x}_0)|\right)|\bar{x}-\bar{x}_0|$$

$$\bar{h}(\bar{x}) - \bar{h}(\bar{x}_0) = D_g(\bar{y}_0)\cdot D_f(\bar{x}_0)(\bar{x}-\bar{x}_0) + \bar{o}(|\bar{x}-\bar{x}_0|)_{\bar{x}\to\bar{x}_0}$$